BidPilot: a simple guide

What the app does, what you see on each page, and where the AI (LLM) fits in

In one sentence: BidPilot is an AI helper for an online shop's ads. Every day it decides how much to pay for each product ad and keyword ad, explains why, and asks you before big changes. Once a week it acts like a consultant: it studies the shop's data and the web and gives a short, ranked to-do list where every number and fact has a source you can click. **Everything runs on a practice ad account** (a fake shop with 40 products and 120 keywords), so no real money is spent.

How to open it

In the project folder run `python -m cli demo` (or `streamlit run app/dashboard.py`), then open **http://localhost:8501** in a browser. The left sidebar shows which simulated day the account is on, whether the AI is connected, and a **Reset account to day 0** button to start over. Right now the AI shows as *off*; that is expected (see page 2).

1. Store

What it shows	The fake shop: 40 home-textile products (bedsheets, curtains, towels, cushions, rugs) with price, profit margin after marketplace fees and returns, stock left and *days of stock*. Colours: yellow = running out soon, red = out of stock, blue = too much stock. At the bottom, **Seller objectives** lets you pick products to push for more sales, with a limit on how expensive the ads may get (an ACOS cap).
Try this	Look for the red and yellow rows: the agent will stop or slow ads for those. Optionally pick one blue (overstocked) product as a growth product and save.

2. Run a day

What it shows	The daily job. **Plan day N** makes the agent study yesterday's results and propose new bids for all 160 ads. The **maths** (not the AI) picks the numbers. Safety rules then: never move a bid more than 20% a day, cut ads for products that are running out, pause keywords that keep spending with zero sales (only when the evidence is strong), and protect the best earners. Big changes (more than 10%, any pause, the top spenders) wait for **your** approval; small ones apply by themselves. Each change has a reason with real numbers, e.g. *"ROAS 4.25 this week vs 1.83 the week before"*, plus a confidence level.
Try this	Click Plan day 1. Approve one change, reject one, then **Apply decisions and run the day**. The day's result appears underneath. **Fast-forward** runs several days with everything approved.

3. 14-day race

What it shows	The main proof. The same fake shop is run for 14 days three ways: **fixed** (nobody touches the bids), **rules** (a simple tool: "losing money? cut 10%") and the **agent**. All three see exactly the same luck, so the comparison is fair. The chart shows profit building up day by day; the table shows sales, ad spend, ACOS/ROAS, stock availability and profit.
Try this	Switch between *Total* profit (all sales, ads + organic) and *Ad-attributed* profit. The agent line ends on top in both.

14 simulated days	Total profit	Profit from ad sales	Ad money wasted on out-of-stock products
fixed (do nothing)	Rs 807,137	Rs 128,820	Rs 9,640
rules (simple tool)	Rs 842,640	Rs 155,987	Rs 10,161
BidPilot agent	**Rs 918,161**	**Rs 220,167**	**Rs 0**

The agent sells about 4% less than "do nothing" but makes the most money: it stops paying for sales that lose money (cheap-margin best-sellers) and never advertises products that are out of stock. These are **simulated** results: they show the method works, not a promise for a real account.

4. Consultant

What it shows	The weekly report (days 7 and 14, or on request). A ranked list of up to 5 recommendations, each with priority, confidence, risk and the next step. Every claim has clickable chips: **[D#]** = a result from our own data (click to see the question, the code that was run and the table), **[W#]** = a fact from a real web page (click to see the link, the exact quote and the date). Anything that cannot be matched to a real source is removed or greyed out as *unverified*. You accept or reject each one: accepted bid ideas go into tomorrow's plan (still through the safety rules); others (restock, change a price) go to your to-do list.
Try this	**Empty for now**: this page needs the AI, which will run on the AMD GPU. Nothing is broken.

5. Ask the data

What it shows	Ask a question about the shop. The AI writes a small piece of code, a safety check approves it, it runs in a locked box (no internet, no files), and the answer comes back with a [D#] source, the code and the result table. Without the AI, six ready-made questions work (wasted keywords, best category, out-of-stock spend, low-margin products, weak product pages, overstock worth pushing).
Try this	Click "Which products get many clicks but convert below their category average?". It finds the three products we planted with weak product pages.

6. Experiments

What it shows	Ad platforms give the ad credit for every sale, but some buyers would have bought anyway. An experiment turns one product's ads **on and off on alternating days** and compares total sales. The result says what share of the ad sales were truly extra, with a range of certainty. If the ads clearly bring fewer extra sales than claimed, the agent bids less on that product. Tests pause ads, so you approve them first.
Try this	Propose the suggested product, Approve, then fast-forward 14 days on Run a day. For our planted best-seller the test shows that only a small share of its ad sales were truly extra.

7. Speed

What it shows	Will compare the slow way (one AI call per ad, one after another, AI guessing bids) with BidPilot's way (maths for all bids in under a second, AI explanations written all at the same time) on the AMD GPU.
Try this	**Empty for now**: it needs a real measurement on the AMD GPU.

About the AI (LLM), in simple words

- **What the LLM is:** the AI language model (an open model, Qwen 32B) that we run ourselves on an AMD GPU using vLLM. It writes text: plans, explanations, analysis code and the consultant report.
- **What it never does:** it never chooses bid amounts (the maths does), and it never touches real money. Its code only runs inside the locked box after a safety check.
- **How we keep it honest:** every number it writes is checked against the real data. If it makes a number up, its sentence is thrown away and a plain, safe sentence is used instead. Web facts are kept only if the exact quote is found on the page.
- **Right now:** no AI server was reachable from this laptop, so the app runs in *no-AI mode*: bidding, the race, the stock rules, experiments and the six ready-made questions all work; explanations use simple fixed sentences; the Consultant and Speed pages wait for the AI. All AI parts were tested with stand-ins and pass, but they have not yet run against the real model.
- **What happens on the AMD box:** we point the app at the AMD server (one address) and run three commands: a quick check that the AI answers correctly, a demo run that fills the Consultant page, and the speed test that fills the Speed page with real numbers.

What is still needed for the final submission

Send me the AMD server address (for example `http://<ip>:8000/v1`) or terminal access to the box. I will: (1) check the AI connection, (2) produce the race with the full AI agent and the day-7 consultant report, (3) run the speed test and save the real numbers, and (4) rehearse the 4-minute demo with you. Until then, everything except the Consultant and Speed pages can be shown today, and the demo works offline.